

Project Document: Advanced Company Network Design

1. Introduction

This document details the design and implementation of an **Advanced Company Network**, created using Cisco Packet Tracer. The project establishes a secure, highly available, and scalable enterprise network infrastructure, integrating multiple services (Data, Voice, Wireless, Cloud connectivity) and advanced Cisco technologies.

2. Project Goals

- Implement a **Hierarchical Network Design** (Core, Distribution, Access) for scalability and fault isolation.
- Establish **robust security** using a Cisco ASA Firewall and inspection policies.
- Ensure network **High Availability (HA)** at the core and access layers.
- Provide segmentation through **VLANs** and efficient **Inter-VLAN Routing**.
- Deploy a professional **VoIP** and **Corporate Wireless** solution.
- Secure remote access to network devices using **SSH** and **ACLs**.

3. Network Design and Addressing

Component	Design Choice	Implementation Details
Topology	Hierarchical Model	Separated Core, Distribution (Layer 3 Switch), and Access Layers.
Addressing	Subnetting	Used a /24 or similar block, logically subnetted to allocate specific ranges for each VLAN
VLANs	Segmentation	VLAN 10 (MGT), 20 (LAN), 50 (WLAN), 70 (VoIP), 90 (SERVERS), 199 (BLACKHOLE).
Routing	OSPF (Internal)	Configured OSPF for dynamic routing between distribution switches and the firewall/router interfaces.

4. Implemented Technologies and Configurations

- ❖ Network Security (Cisco ASA Firewall)
 - **Zones:** Defined **Inside**, **Outside**, and **DMZ** security zones.
 - **NAT:** Configured **PAT (Port Address Translation)** for Inside users accessing the Internet and **Static NAT** for server services in the DMZ.
 - **ACLs:** Applied inspection policies (Access Control Lists) to regulate traffic between zones
- ❖ High Availability and Redundancy
 - **HSRP:** Implemented on **Core/Distribution Switches** to create a virtual gateway IP, ensuring default gateway redundancy for end-user VLANs
 - **LACP EtherChannel:** Used **LACP (Link Aggregation Control Protocol)** to bundle physical links between the Core and Distribution layers, providing fault tolerance and increased bandwidth.
- ❖ Switching and Layer 2 Features
 - **Inter-VLAN Routing:** Achieved using **SVI (Switched Virtual Interfaces)** on the Layer 3 Distribution switch.
 - **STP Enhancements:** Configured **STP PortFast** on all access ports to ensure immediate link activation, and **BPDUGuard** to protect the spanning-tree topology from unauthorized devices.
- ❖ Services
 - **Wireless:** Deployed a **Cisco Wireless LAN Controller (WLC)** managing **Lightweight Access Points (LAPs)**. Configured multiple SSIDs for segmentation (**Corporate, Guest, Employees**)
 - **Telephony:** Integrated **Cisco Voice Gateways** and **IP Phones**. Configured a voice VLAN and DHCP options for phone provisioning.
 - **DHCP: Active Directory** (simulated server) configured as the main DHCP server for centralized IP management.

5. Connectivity and Verification

- **ISP/AWS Integration:** Simulated external connectivity to an ISP and an AWS cloud resource using a Default Static Route (`ip route 0.0.0.0 0.0.0.0 ...`).
- **Verification:** Tested end-to-end connectivity using `ping`, `tracert`, and verified security policies were in effect by attempting blocked connections.

6. Conclusion

This project demonstrates proficiency in securing, configuring, and maintaining a modern enterprise network using a wide array of advanced Cisco IOS features and best practices.

IP Tables

Department/Area	Network Address	Usable IP Range (From - To)	Gateway IP	Broadcast IP
Management	192.168.10.0/24	192.168.10.1 - 192.168.10.254	192.168.10.1	192.168.10.255
WLAN	10.20.0.0/16	10.20.0.1 - 10.20.255.254	10.20.0.1	10.20.255.254
LAN	172.16.0.0/16	172.16.0.1 - 172.16.255.254	172.16.0.1	172.16.255.255
VOIP	172.30.0.0/16	172.30.0.1 - 172.30.255.254	172.30.0.1	172.30.255.255
DMZ	10.11.11.0/27	10.11.11.1 - 10.11.11.30	10.11.11.1	10.11.11.31
INSIDE SERVERS	10.11.11.32/27	10.11.11.33 - 10.11.11.62	10.11.11.33	10.11.11.63

No.	Network Address
CLOUD Area	8.0.0.0/8
ISP 1 - Internet	20.20.20.0/30
ISP 2 - Internet	30.30.30.0/30
ISP 1 - FWL1	105.100.50.0/30
ISP 2 - FWL2	105.100.50.4/30
ISP 1 - FWL1	205.200.100.0/30
ISP 2 - FWL2	205.200.100.4/30
FWL1 to MLSW1	10.2.2.0/30
FWL2 to MLSW2	10.2.2.4/30
FWL1 to MLSW1	10.2.2.8/30
FWL2 to MLSW2	10.2.2.12/30